Gray Reid

COMPUTATIONAL MODELING AND MACHINE LEARNING, NYC

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Profile

PhD graduate in computational physics with expertise in machine learning and scientific computing, specializing in algorithm development for large-scale simulations. Proven track record in enhancing computational efficiency and accuracy over state-of-the-art techniques. Eager to transition into an industry ML role to apply advanced modeling skills and drive innovation in tech. Committed to leveraging strong research background to contribute to high-impact projects.

Skills

Machine Learning

- Neural Networks, Deep Learning, Reinforcement Learning
- Attention Mechanisms, Transformers, CNNs, RNNs, GANs, Diffusion Models, GRUs, LSTMs

Scientific Computing

- HPC, Numerical Methods, Computational Modeling
- Finite Difference, Finite Element, Finite Volume, Spectral Methods, Monte Carlo, AMR, Multigrid
- Optimization, Profiling, Debugging

Quantitative Methods

- Calculus, PDEs, SDEs, ODEs, Linear Algebra, Probability, Statistics, Signal Processing
- Lagrangian Mechanics, Hamiltonian Mechanics
- Quantum Mechanics, Statistical Mechanics, Thermodynamics
- Optics, Spectroscopy, Computational Optics
- Field Theory, Fluid Dynamics, Electromagnetism, General Relativity

Languages and Frameworks

- C, Python, Fortran
- Pytorch, Jax, TensorFlow, Pandas
- MPI, OpenMP, CUDA

Numerical Analysis

- Error analysis, Stability Analysis, Complexity Analysis, Convergence Analysis
- Conditioning, Discretization

Research

- Research Design, Scientific Writing, Literature Review
- Data Analysis, Data Visualization
- Mentoring, Curriculum Development, Presentations

Select Awards

NSERC CGSD
Canada's premier doctoral fellowship — *UBC*

Four Year Fellowship
Doctoral funding based on academic excellence — *UBC*

NSERC CGSM
Canada's premier master's fellowship — *UBC*

Governor General's Medal
Canada's highest academic honor — *Acadia*

Education

PhD Physics — *Vancouver, BC, Canada*
UNIVERSITY OF BRITISH COLUMBIA — *2024*

- Improved resolution and accuracy of high performance simulations of gravitational collapse by orders of magnitude over state-of-the-art without significantly increasing computational cost
- Developed expertise in computational modeling, numerical methods, high performance computing, debugging and optimization of massively parallel systems

BSc Honours Physics — *Wolfville, NS, Canada*
ACADIA UNIVERSITY — *2012*

- Highlighted deficiencies in existing techniques for modeling TEM imaging of materials and developed new methods for addressing these deficiencies
- Developed improved equations describing cavity ring-down spectroscopy for use in chemical sensing applications

Select Publications

Universality in the Critical Collapse of the Einstein-Maxwell System
PHYSICAL REVIEW D — *2023*

- Improved resolution and accuracy by several orders of magnitude over state of the art
- Improved resolution and accuracy of in-lab adaptive mesh refinement (AMR) software for simulation of generic PDEs
- Resolved conflicts with previous investigations and provided a framework for understanding discrepancies in related literature.

Reference Metric Approach to the Z4 System
PHYSICAL REVIEW D — *2023*

- Achieved performance equivalent or exceeding leading formulations of General Relativity such as GBSSN and FCCZ4
- Derived a novel hyperbolic formulation of General Relativity
- Demonstrated the stability of the formulation through pseudodifferential first order reduction and simulations in the strong field regime

Stability of Nonminimally Coupled Topological-Defect Boson Stars
PHYSICAL REVIEW D — *2024*

- Resolved the open question of stability for topological-defect boson stars in modified gravity
- Performed comprehensive non-linear and semi-analytic stability analysis for large families of solutions